

Sporadic acute or fulminant hepatitis E in Hokkaido, Japan, may be food-borne, as suggested by the presence of hepatitis E virus in pig liver as food.

Among ten patients who contracted sporadic acute or fulminant hepatitis E between 2001 and 2002 in Hokkaido, Japan, nine (90 %) had a history of consuming grilled or undercooked pig liver 2-8 weeks before the disease onset. We tested packages of raw pig liver sold in grocery stores as food in Hokkaido for the presence of hepatitis E virus (HEV) RNA by RT-PCR. Pig liver specimens from seven (1.9 %) of 363 packages had detectable HEV RNA. Partial sequence analyses revealed that the seven swine HEV isolates belonged to genotype III or IV. One swine HEV isolate (swJL145) from a packaged pig liver had 100 % identity with the HE-JA18 isolate recovered from an 86-year-old patient in Hokkaido. Two swine HEV isolates (swJL234 and swJL325) had 98.5-100 % identity with the HE-JA4 isolate obtained from a 44-year-old patient in Hokkaido. These results indicate that inadequately cooked pig liver may transmit HEV to humans.